

# Lecture 9: Hypothesis Testing and Confidence Intervals

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## 1. Agenda and Announcements

- Today

- Hypothesis Testing and Confidence Intervals
- Top Hat Quiz

- Next Class

- Article Discussion: Shirani-Mehr, Houshmand, David Rothschild, Sharad Goel, and Andrew Gelman. 2018. "Disentangling bias and variance in election polls." *Journal of the American Statistical Association* 113(522): 607-614.

## 2. Hypothesis testing

### 2.1 What is a hypothesis

- A falsifiable statement about what we believe will happen based on the theory we are trying to test.

### 2.2 Falsifiable:

- We start from the assumption that our theory is wrong
- Our hypothesis is called the

**alternative hypothesis** or **H1**

- because it is the alternative to our assumption of no relationship

**null hypothesis** or **H0**

### 2.3 Testing the null hypothesis

- Statistical tests show the degree of certainty that we can reject the null hypothesis
- They show the probability that the alternative hypothesis is due to random chance
- When the probability,  $p$ , is below our pre-determined threshold, we reject the null hypothesis
- **If we reject the null hypothesis, the alternative hypothesis is true, right?**

### 2.4 NO!

- **If we reject the null hypothesis, the alternative hypothesis is true, right?**

**NO!!!!**

- If we reject the null, we infer that the alternative hypothesis (our hypothesis) is approximately true within the probability we have chosen.
- If we reject the null, we can say there is strong evidence that the alternative hypothesis (our hypothesis) is true
- We often test multiple hypotheses

## 2.5 How do we get to testing the null?

**How do we get from a sample with a correlation to talking about testing a hypothesis for a population?**

- Tying sample statistics to population parameters (LLN)
- Probability distributions
- Tying our data to the probability distributions (CLT)

## 2.6 Statistics to parameters

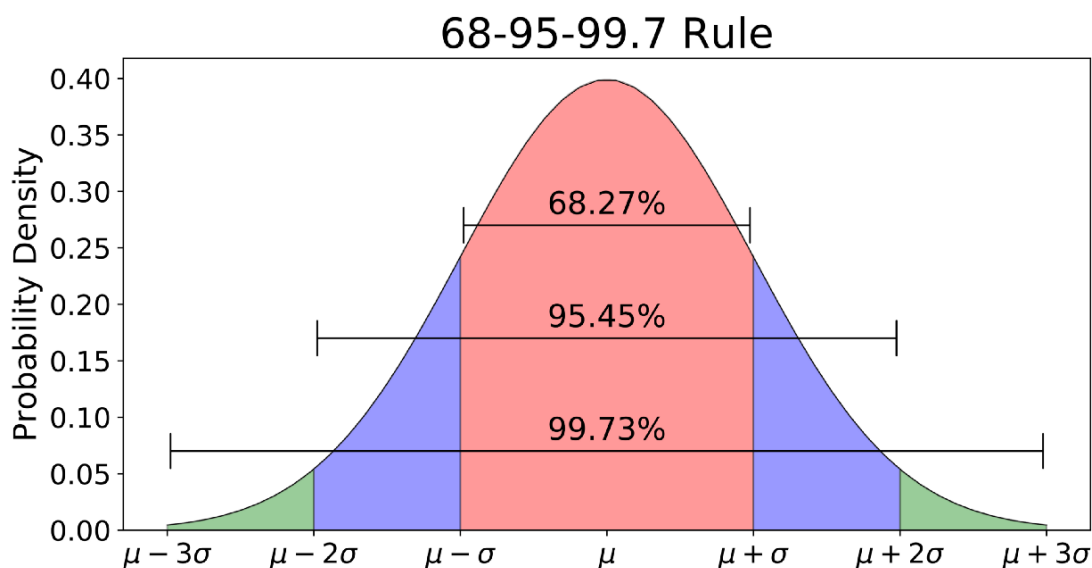
- We rely on the Law of Large Numbers and use sample statistics
- With sufficient sample size and proper math, our sample mean is an unbiased estimator of the population mean
- With CLT,  $s^2$  is an unbiased and consistent estimator of  $\sigma^2$  - For the normal distribution  $s^2$  is also the Minimum Variance Unbiased Estimator for  $\sigma^2$

## 2.7 Probability distributions

The 68-95-99.7 Rule

- Allows us to estimate probability based on distance from the mean
- Applies to normal distribution
- Basis for the actual decision rules

## 2.8 The 68-95-99.7 Rule



68-95-99.7 rule

## 2.9 From Distribution to Confidence

- We can now bridge the gap between probability distributions and our sample data.
- CLT ties our sample means to the normal distribution
- Sample stats are unbiased estimators of population parameters
- They are only *point estimates*, so we construct a *confidence interval*

- *\*confidence interval\** a range of values around the *\*point estimate\** that we are X% confident contains the true population parameter

## 2.10 Math and philosophy of the confidence interval

- Math and philosophy detail: The true population parameter is a fixed but unknown number, while our calculated interval is random but known with certainty. Our X percent confidence reflects the long-run reliability of our method. If we repeated this exact sampling process 100 times, X of the resulting intervals would successfully capture the true population parameter.
- 

## 2.11 Calculating the Interval

- we need three pieces of information:
- point estimate = sample statistics
- standard error = the standard deviation of the sampling distribution
- critical value derived from our desired confidence level - for normal distribution this is a z-score
- For 95% confidence, the critical z is 1.96

## 2.12 Confidence Interval of the Mean

- The general formula for a confidence interval of the mean is:

$$CI = \bar{x} \pm z \left( \frac{s}{\sqrt{n}} \right)$$

## 2.13 Confidence Interval of the variance

- More complicated formula

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{1-\alpha/2}}$$

- Because we have to use the Chi Square,  $\chi^2$ , distribution
- We will discuss  $\chi^2$  more in hypothesis testing

## 2.14 CI of the standard deviation

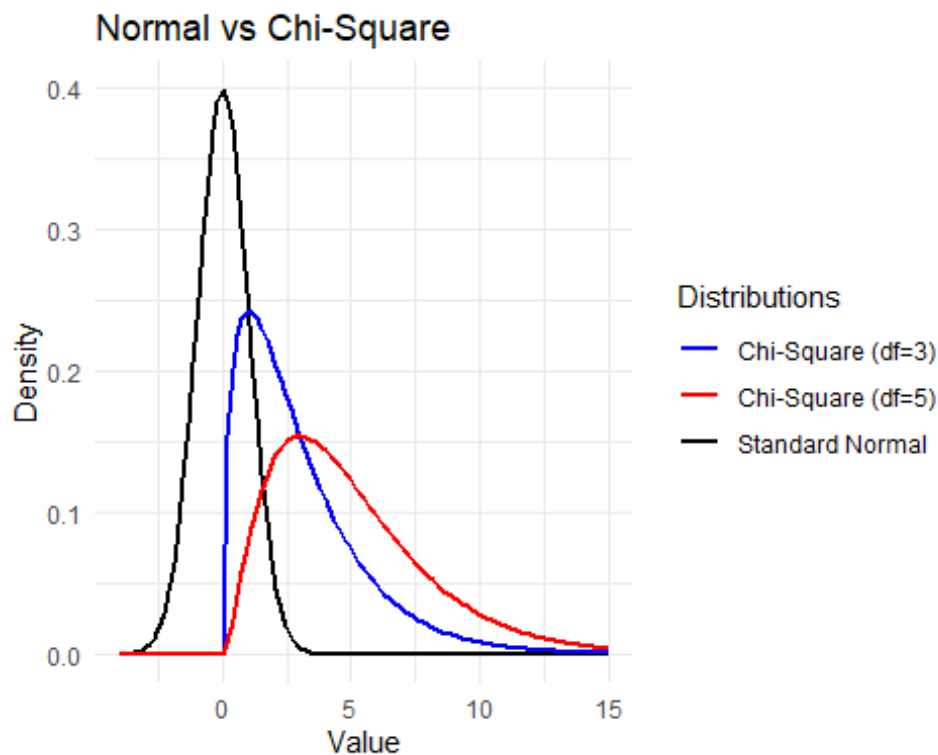
$$\sqrt{\frac{(n-1)s^2}{\chi_{\alpha/2}^2}} \leq \sigma \leq \sqrt{\frac{(n-1)s^2}{\chi_{1-\alpha/2}^2}}$$

- Also uses the  $\chi^2$  distribution

## 2.15 Why the $\chi^2$ not the normal?

- What did we do the deviations from the mean,  $(x_i - \bar{x})$  to compute variance?
- When we square distances from the mean to get the variance, the distribution changes
- By eliminating all negative numbers, the distribution becomes skewed and looks like this:

## 2.16 Distribution Comparison



## 2.17 Authorship, License, Credits

- [Stephen Moore](#) code used to simulate the Law of Large Numbers

- 68-95-99 rule graphic [Source:https://towardsdatascience.com/understanding-the-68-95-99-7-rule-for-a-normal-distribution-b7b7cbf760c2](https://towardsdatascience.com/understanding-the-68-95-99-7-rule-for-a-normal-distribution-b7b7cbf760c2)
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